$$C_{\phi q}^{(22)[1]\,i1\_i2\_} \to$$

$$\frac{1}{16\pi^2}\ \Big(\frac{1}{81}\sum_p g_1^4\, LF_{3,0}[m_{\tilde{d}}^{p}]\,\delta_{i1i2} - \frac{5}{216}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde{d}}^{p}]\,\delta_{i1i2} + \frac{4}{405}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde{d}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{27}\sum_p g_1^4\, LF_{3,0}[m_{\tilde{e}}^{p}]\,\delta_{i1i2} - \frac{5}{72}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde{e}}^{p}]\,\delta_{i1i2} + \frac{4}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde{e}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{54}\sum_p g_1^4\, LF_{3,0}[m_{\tilde{l}}^{p}]\,\delta_{i1i2} - \frac{5}{144}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde{l}}^{p}]\,\delta_{i1i2} + \frac{2}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde{l}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{162}\sum_p g_1^4\, LF_{3,0}[m_{\tilde{q}}^{p}]\,\delta_{i1i2} - \frac{5}{432}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \frac{2}{405}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde{q}}^{p}]\,\delta_{i1i2} +$$

$$\frac{4}{81}\sum_p g_1^4\, LF_{3,0}[m_{\tilde{u}}^{p}]\,\delta_{i1i2} - \frac{5}{54}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde{u}}^{p}]\,\delta_{i1i2} + \frac{16}{405}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde{u}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{54} g_1^4\, LF_{3,0}[\tilde{\mu}]\,\delta_{i1i2} + \frac{1}{36} g_1^4\, LF_{4,-1}[\tilde{\mu}]\,\delta_{i1i2} - \frac{4}{135} g_1^4\, LF_{5,-2}[\tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{18} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,0}[m_1, m_{\tilde{d}}^{p}] - \frac{1}{9} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,-1}[m_1, m_{\tilde{d}}^{p}] +$$

$$\frac{1}{18} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{4,1,-2}[m_1, m_{\tilde{d}}^{p}] + \frac{1}{2592} g_1^4\, LF_{2,1,0}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{2,2,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} - \frac{1}{1296} g_1^4\, LF_{3,1,-1}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{4,1,-2}[m_1, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{1}{2592} g_1^4\, LF_{2,1,0}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{2,2,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} - \frac{1}{1296} g_1^4\, LF_{3,1,-1}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{4,1,-2}[m_1, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, LF_{2,1,0}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{2,2,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2\, LF_{3,1,-1}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{4,1,-2}[m_2, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2\, LF_{2,1,0}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{2,2,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2\, LF_{3,1,-1}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{4,1,-2}[m_2, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} + \frac{2}{3} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,0}[m_3, m_{\tilde{d}}^{p}] -$$

$$\frac{4}{3} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,-1}[m_3, m_{\tilde{d}}^{p}] + \frac{2}{3} g_3^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{4,1,-2}[m_3, m_{\tilde{d}}^{p}] +$$

$$\frac{1}{54} g_1^2 g_3^2\, LF_{2,1,0}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2\, LF_{2,2,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} -$$

$$\frac{1}{27} g_1^2 g_3^2\, LF_{3,1,-1}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2\, LF_{4,1,-2}[m_3, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} +$$

$$\frac{1}{54} g_1^2 g_3^2\, LF_{2,1,0}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2\, LF_{2,2,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} -$$

$$\frac{1}{27} g_1^2 g_3^2\, LF_{3,1,-1}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} + \frac{1}{54} g_1^2 g_3^2\, LF_{4,1,-2}[m_3, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} +$$

$$\frac{1}{18} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,0}[m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{1}{36} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,2,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}] -$$

$$\frac{1}{36} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,-1}[m_{\tilde{d}}^{p}, \tilde{\mu}] + \frac{1}{18} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} -$$

$$\frac{1}{36} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,2,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \frac{1}{18} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}]\,\delta_{i1i2} -$$

$$\frac{1}{36} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,2,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}]\,\delta_{i1i2} - \frac{1}{18} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} -$$

$$\frac{1}{18} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} + \frac{1}{8} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} -$$

$$\frac{1}{18} g_1^2\, \overline{a_e}^{pr} a_e^{pr}\, LF_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}]\,\delta_{i1i2} - \frac{1}{18} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} -$$

$$\frac{1}{18} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} + \frac{5}{24} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} -$$

$$\frac{1}{6} g_1^2\, \overline{a_d}^{pr} a_d^{pr}\, LF_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}]\,\delta_{i1i2} - \frac{1}{36} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2} +$$

$$\frac{1}{72} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}]\,\delta_{i1i2} - \frac{1}{1296} g_1^4\, LF_{2,1,0}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{3,1,-1}[m_{\tilde{q}}^{i1}, m_1]\,\delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2\, LF_{2,1,0}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{3,1,-1}[m_{\tilde{q}}^{i1}, m_2]\,\delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2\, LF_{2,1,0}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2} +$$

$$\frac{1}{54} g_1^2 g_3^2\, LF_{3,1,-1}[m_{\tilde{q}}^{i1}, m_3]\,\delta_{i1i2} - \frac{1}{1296} g_1^4\, LF_{2,1,0}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2} +$$

$$\frac{1}{2592} g_1^4\, LF_{3,1,-1}[m_{\tilde{q}}^{i2}, m_1]\,\delta_{i1i2} - \frac{1}{48} g_1^2 g_2^2\, LF_{2,1,0}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2} +$$

$$\frac{1}{96} g_1^2 g_2^2\, LF_{3,1,-1}[m_{\tilde{q}}^{i2}, m_2]\,\delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2\, LF_{2,1,0}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2} +$$

$$\frac{1}{54} g_1^2 g_3^2\, LF_{3,1,-1}[m_{\tilde{q}}^{i2}, m_3]\,\delta_{i1i2} - \frac{1}{9} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,0}[m_{\tilde{u}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{18} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,2,-1}[m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{1}{18} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{3,1,-1}[m_{\tilde{u}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{36} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} + \frac{1}{36} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} +$$

$$\frac{1}{12} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} - \frac{1}{6} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{pr} y_u^{pr}\, LF_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}]\,\delta_{i1i2} -$$

$$\frac{5}{72} g_1^4\, LF_{4,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} + \frac{1}{18} g_1^4\, LF_{5,1,-3}[\tilde{\mu}, m_1]\,\delta_{i1i2} -$$

$$\frac{5}{24} g_1^2 g_2^2\, LF_{4,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} + \frac{1}{6} g_1^2 g_2^2\, LF_{5,1,-3}[\tilde{\mu}, m_2]\,\delta_{i1i2} -$$

$$\frac{1}{36} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,0}[\tilde{\mu}, m_{\tilde{d}}^{p}] - \frac{5}{72} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,-1}[\tilde{\mu}, m_{\tilde{d}}^{p}] +$$

$$\frac{1}{72} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{4,1,-2}[\tilde{\mu}, m_{\tilde{d}}^{p}] + \frac{1}{2}\, \overline{y_d}^{rp}\, \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p}\, LF_{2,1,0}[\tilde{\mu}, m_{\tilde{q}}^{r}] -$$

$$\overline{y_d}^{rp}\, \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p}\, LF_{3,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{r}] + \frac{1}{2}\, \overline{y_d}^{rp}\, \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p}\, LF_{4,1,-2}[\tilde{\mu}, m_{\tilde{q}}^{r}] +$$

$$\frac{1}{18} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,0}[\tilde{\mu}, m_{\tilde{u}}^{p}] + \frac{1}{72} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{3,1,-1}[\tilde{\mu}, m_{\tilde{u}}^{p}] +$$

$$\frac{1}{72} g_1^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{4,1,-2}[\tilde{\mu}, m_{\tilde{u}}^{p}] + \frac{1}{72} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] +$$

$$\frac{1}{144} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,2,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] - \frac{1}{72} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{3,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] +$$

$$\frac{1}{72} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{1}{144} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,2,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] -$$

$$\frac{1}{72} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{3,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] - \frac{1}{4} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{6} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,1,-2}[m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{1}{144} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{p}] +$$

$$\frac{1}{288} g_1^4\, LF_{2,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2} + \frac{1}{144} g_1^4 m_1^2\, LF_{2,1,1,0}[m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2} -$$

$$\frac{1}{144} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}] + \frac{1}{288} g_1^4\, LF_{2,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{1}{144} g_1^4 m_1^2\, LF_{2,1,1,0}[m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2} + \frac{7}{48} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}^{p}] +$$

$$\frac{1}{288}\left(6 g_1^2\, \overline{y_d}^{i2p} y_d^{i1p} + g_1^4\,\delta_{i1i2}\right) LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{288} g_1^4 m_1^2\, LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{1}{288}\left(6 g_1^2\, \overline{y_d}^{i2p} y_d^{i1p} + g_1^4\,\delta_{i1i2}\right)$$

$$LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}] - \frac{1}{288} g_1^4 m_1^2\, LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} -$$

$$\frac{1}{16} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{u}}^{p}] + \frac{3}{32} g_2^4\, LF_{2,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{3}{16} g_2^4 m_2^2\, LF_{2,1,1,0}[m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}]\,\delta_{i1i2} + \frac{3}{32} g_2^4\, LF_{2,1,1,-1}[m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2} +$$

$$\frac{3}{16} g_2^4 m_2^2\, LF_{2,1,1,0}[m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}]\,\delta_{i1i2} + \frac{3}{16} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{d}}^{p}] +$$

$$\frac{3}{32}\left(2 g_2^2\, \overline{y_d}^{i2p} y_d^{i1p} + g_2^4\,\delta_{i1i2}\right) LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] - \frac{3}{32} g_2^4 m_2^2$$

$$LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}]\,\delta_{i1i2} + \frac{3}{32}\left(2 g_2^2\, \overline{y_d}^{i2p} y_d^{i1p} + g_2^4\,\delta_{i1i2}\right) LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}] -$$

$$\frac{3}{32} g_2^4 m_2^2\, LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i2}]\,\delta_{i1i2} - \frac{3}{16} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{u}}^{p}] -$$

$$\frac{1}{3} m_3 g_3^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,0}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] - \frac{1}{6} m_3 g_3^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,2,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] +$$

$$\frac{1}{3} m_3 g_3^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{3,1,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] - \frac{1}{3} m_3 g_3^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,0}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] -$$

$$\frac{1}{6} m_3 g_3^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,2,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{1}{3} m_3 g_3^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{3,1,1,-1}[m_3, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] +$$

$$\frac{1}{6} m_3 g_3^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{d}}^{p}] + \frac{1}{6} m_3 g_3^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i2}, m_{\tilde{d}}^{p}] -$$

$$\frac{1}{16} \tilde{\mu}^2\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} y_u^{ps}\, LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{u}}^{s}] - \frac{1}{16} \tilde{\mu}^2\, \overline{y_d}^{i2p} y_d^{rp}\, \overline{y_u}^{rs} y_u^{i1s}$$

$$LF_{2,2,1,-1}[m_{\tilde{q}}^{r}, \tilde{\mu}, m_{\tilde{u}}^{s}] + \frac{1}{16} \tilde{\mu}^2\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} y_u^{ps}\, LF_{2,2,1,-1}[m_{\tilde{u}}^{s}, \tilde{\mu}, m_{\tilde{q}}^{p}] +$$

$$\frac{1}{16} \tilde{\mu}^2\, \overline{y_d}^{i2p} y_d^{rp}\, \overline{y_u}^{rs} y_u^{i1s}\, LF_{2,2,1,-1}[m_{\tilde{u}}^{s}, \tilde{\mu}, m_{\tilde{q}}^{r}] + \frac{1}{16} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}] -$$

$$\frac{1}{16} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] + \frac{1}{24} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}] -$$

$$\frac{1}{16} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}] + \frac{1}{24} g_1^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}] +$$

$$\frac{1}{8} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{u}}^{p}] + \frac{3}{16} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}] -$$

$$\frac{9}{16} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] + \frac{3}{8} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}] -$$

$$\frac{9}{16} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}] + \frac{3}{8} g_2^2\, \overline{y_d}^{i2p} y_d^{i1p}\, LF_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}] +$$

$$\frac{3}{8} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{u}}^{p}] - \frac{1}{8} \tilde{\mu}^2\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} y_u^{ps}\, LF_{2,1,1,0}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}] +$$

$$\frac{1}{8} \tilde{\mu}^2\, \overline{y_d}^{pr} y_d^{i1r}\, \overline{y_u}^{i2s} y_u^{ps}\, LF_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{s}] - \frac{1}{8} \tilde{\mu}^2\, \overline{y_d}^{i2p} y_d^{rp}\, \overline{y_u}^{rs} y_u^{i1s}$$

$$LF_{2,1,1,0}[\tilde{\mu}, m_{\tilde{q}}^{r}, m_{\tilde{u}}^{s}] + \frac{1}{8} \tilde{\mu}^2\, \overline{y_d}^{i2p} y_d^{rp}\, \overline{y_u}^{rs} y_u^{i1s}\, LF_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{q}}^{r}, m_{\tilde{u}}^{s}] -$$

$$\frac{1}{48} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{96} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{48} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{1,1,1,1,0}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] -$$

$$\frac{1}{96} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{48} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] -$$

$$\frac{1}{48} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{1,1,1,1,0}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{16} m_2 g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] -$$

$$\frac{3}{32} m_2 g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{3}{16} m_2 g_2^2\, y_d^{i1p}\, \overline{a_d}^{i2p}$$

$$LF_{1,1,1,1,0}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{3}{32} m_2 g_2^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}, \tilde{\mu}] +$$

$$\frac{3}{16} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}, \tilde{\mu}] + \frac{3}{16} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] -$$

$$\frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{1}{144} g_1^2\, \overline{a_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{1}{96} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{96} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{3}{16} g_2^2\, \overline{a_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{3}{32} m_2 g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] +$$

$$\frac{3}{32} m_2 g_2^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}] - \frac{1}{3} g_3^2\, \overline{a_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{d}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{1}{4}\, \overline{y_d}^{i2s} y_d^{i1r}\, \overline{a_d}^{pr} a_d^{ps}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{s}, \tilde{\mu}] -$$

$$\frac{1}{4} \tilde{\mu}^2\, \overline{y_u}^{rp}\, \overline{y_u}^{i2s} y_u^{rs} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{r}, m_{\tilde{u}}^{p}, m_{\tilde{u}}^{s}, \tilde{\mu}] - \frac{1}{96} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] + \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] -$$

$$\frac{3}{32} m_2 g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i1}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{1}{96} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{d}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_1, m_{\tilde{u}}^{p}, \tilde{\mu}] - \frac{3}{32} m_2 g_2^2\, y_d^{i1p}\, \overline{a_d}^{i2p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{d}}^{p}, \tilde{\mu}] - \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{q}}^{i2}, m_2, m_{\tilde{u}}^{p}, \tilde{\mu}] +$$

$$\frac{1}{144} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] - \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i1}, \tilde{\mu}] - \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_1, m_{\tilde{q}}^{i2}, \tilde{\mu}] +$$

$$\frac{3}{16} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i1}, \tilde{\mu}] + \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_2, m_{\tilde{q}}^{i2}, \tilde{\mu}] +$$

$$\frac{1}{3} g_3^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[m_{\tilde{u}}^{p}, m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{i2}] + \frac{1}{96} m_1 g_1^2\, \overline{y_d}^{i2p} a_d^{i1p}$$

$$LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] + \frac{1}{96} m_1 g_1^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] -$$

$$\frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}] - \frac{1}{96} g_1^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}] + \frac{3}{32} m_2 g_2^2\, \overline{y_d}^{i2p} a_d^{i1p}\, LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i1}] +$$

$$\frac{3}{32} m_2 g_2^2\, y_d^{i1p}\, \overline{a_d}^{i2p}\, LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{p}, m_{\tilde{q}}^{i2}] + \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}$$

$$LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{p}] + \frac{3}{32} g_2^2\, \tilde{\mu}^2\, \overline{y_u}^{i2p} y_u^{i1p}\, LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}^{i2}, m_{\tilde{u}}^{p}]\Big)$$